

Recipe Data Pipeline

Python - BeautifulSoup - Gemini via LiteLLM - github.com/finchteck/scraping

From messy recipe pages to validated CSV + JSON

Problem: recipe content buried in site chrome (nav, ads, footers) with inconsistent markup. Goal: a polite, repeatable pipeline that turns raw HTML into structured data a client can open in Excel or load as JSON.

Approach

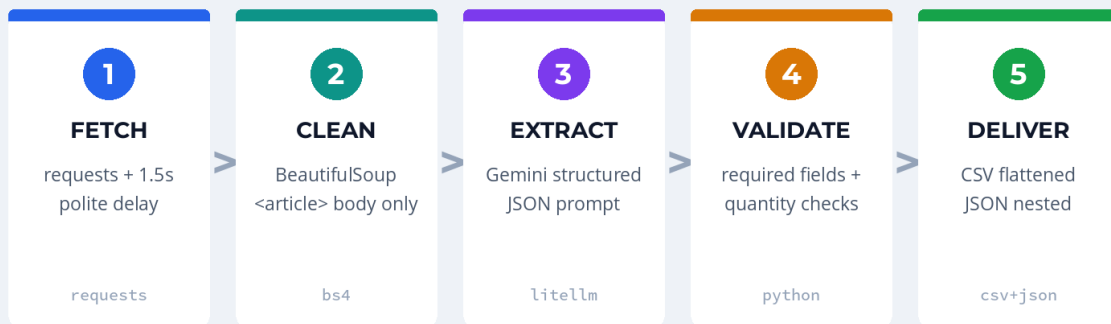
1. **Fetch** with requests + 1.5 s polite delay 2. **Clean** to the <article> body with BeautifulSoup 3. **Extract** via a fixed-schema Gemini prompt 4. **Validate** fields + numeric quantities 5. **Deliver** flattened CSV and nested JSON from one object. No API key? The deterministic fallback produces the same formats with zero setup.

Results

5 recipes processed into **54 validated ingredient rows** (7-19 per recipe). Fractions parsed (1/2 -> 0.5), attached units split (350g -> 350.0 g), subsection headers skipped. One command reproduces everything: `pip install -r requirements.txt && python scraper.py`

Recipe Data Pipeline

HTML > Structured CSV + JSON - Python - BeautifulSoup - Gemini via LiteLLM



RESULTS

5 recipes -> 54 validated ingredient rows -> output.csv + output.json

Zero setup: no API keys, no proxy. Runs anywhere with pip install.

Polite fetching - Token-efficient - Validated output

5 recipes processed

Fractions + attached units parsed

Dual output: output.csv + output.json

Upwork portfolio - Python data demo

Pipeline overview: five stages from fetch to dual-format delivery.

Messy HTML In > Clean Data Out

Before / After - 30-Minute Chicken Curry - validated quantities + dual format

BEFORE - raw HTML

```
<article>
  <h1>30-Minute Chicken Curry</h1>
  <div class="meta">Serves 4 ...
  <nav>Home | Recipes | ...</nav>
  <header>... ads ...</header>
  <ul>
    <li>2 tbsp rapeseed oil</li>
    <li>4 garlic cloves ...</li>
    <li>350g passata</li>
    <li>1 tsp garam masala ...</li>
    <li>1/2 tsp turmeric ...</li>
    ... 20+ noisy nodes ...
  </ul>
  <footer>cookies | signup ...
</article>
```

X nav / ads / footer stripped



AFTER - structured JSON + CSV

```
{ "name": "30-Minute Chicken Curry",
  "ingredients": [
    {"name": "rapeseed oil",
      "quantity": 2.0, "unit": "tbsp"},
    {"name": "passata",
      "quantity": 350.0, "unit": "g"},
    {"name": "ground turmeric",
      "quantity": 0.5, "unit": "tsp"}, ... ],
  "steps": ["Heat oil ...", "Stir ..."],
  "prep_time": null }
```

```
recipe_name, ingredient, qty, unit
Chicken Curry, rapeseed oil, 2.0, tbsp
19 ingredients x 5 recipes = 54 rows
```

BeautifulSoup article select - fixed-schema prompt - 1/2 -> 0.5 CSV + JSON from one validated object

Before / after: raw HTML in, validated JSON + CSV out (Chicken Curry example).